# VARUNA 2047: National Earth Observation Agricultural Resilience

District Evidence Dossier: **Warangal (Telangana)** | Evaluation Year: 2024 | Report ID: VARUNA_WAR_2024_452cdc

**Executive Summary:** This dossier integrates authentic NOAA Climate Prediction Center ONI indices, NASA POWER historical agroclimatological records (1981–2024), and Copernicus Sentinel-2 calibrated NDVI observations for Warangal district. Based on pre-monsoon observations (issuance date: May 31), the validated predictive engine estimates a **42.6% probability** of seasonal monsoon rainfall deficit (<80% of normal). Risk category: **Moderate Deficit Risk**.

## 1. District Agro-Climatic & Exposure Profile

| | | | |
|---|---|---|---|
| **District Name** | Warangal | **Agro-Climatic Zone** | Central Telangana Zone |
| **Centroid Coordinates** | 17.9689°N, 79.5941°E | **Elevation (m AMSL)** | 280 m |
| **Normal Kharif Rain (JJAS)** | 845.8 mm | **Deficit Threshold (<80%)** | 676.6 mm |
| **Net Sown Area** | 158,000 ha | **Irrigation Coverage** | 49.6% |
| **Major Kharif Crops** | Cotton, Paddy, Chilli, Red Gram | **Vulnerability Profile** | Rainfed Cotton / Paddy tract |

## 2. Pre-Monsoon Deficit Risk Estimation (Observable May 31)

| Metric | Engine Output | Interpretation |
|---|---|---|
| Estimated Deficit Probability | **42.6%** | Moderate Deficit Risk |
| 90% Confidence Interval | [34.6% - 50.6%] | Empirical bootstrap error bound |
| Benchmarked Architecture | ENSO-Only Logistic Regression (Validated) | Out-of-sample ROC-AUC: 0.7830 |
| Vegetation Condition Index (VCI) | 80.4 / 100 | Canopy moisture status (Sentinel-2) |

## 3. Evidence-Based Agricultural Resilience Actions

**Primary Advisory:** Monitor onset timeliness and maintain reservoir storage buffers; promote integrated moisture conservation.

- **Ground Truthing:** Conduct mandal-level verification of farm pond storage and soil moisture in un-irrigated zones.
- **Contingency Sowing:** Prepare agricultural extension seeds for short-duration pulses (Red Gram, Greengram) if monsoon onset is delayed.
- **Water Resource Allocation:** Schedule minor irrigation tank filling from Godavari canal distributaries.
- **Micro-Irrigation Prioritization:** Inspect drip/sprinkler subsidies under PMKSY in tail-end commands.

## 4. Data Provenance & Scientific Integrity Statement

**Data Sources:** Oceanic Niño Index (NOAA Climate Prediction Center, ERSST.v5). Precipitation: NASA POWER Agroclimatology (0.5° gridded daily/monthly). Vegetation: Copernicus Sentinel-2 MSI Band 4/8 and MODIS Terra. Cartography: ISRO NRSC Bhuvan Open Web Map Service (WMS).
**Methodological Limitations:** Deficit probability reflects statistical association based on 1982–2024 climate records. Teleconnection anomalies do not guarantee identical local impacts. This report provides decision support and does not replace official advisories issued by the India Meteorological Department (IMD) or Telangana State Department of Agriculture.